

Abstract No.

Cannabinoid administration as a therapeutic strategy to improve Glioblastoma response to radiotherapy

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Glioblastoma (GBM) is the most frequent brain primary tumor and one of the most aggressive cancer types. Standard treatment consists on combined administration of radiotherapy (RT) and temozolomide (TMZ), although GBMs are **highly resistant to radio-chemotherapy** and **patients frequently recur** after its administration. These features are partly attributed to **Glioma Initiating Cells (GICs)**, a stem-like tumor cell population that promotes tumor maintenance and relapse.

Cannabinoids, the main active compounds of marijuana, are being investigated as potential therapeutic agents. Previous studies from our group showed that **THC (delta-9-tetrahydrocannabinol)**, alone or combined with **CBD (cannabidiol)**, together with TMZ targets GICs *in vitro* and strongly **reduces GBM xenograft growth *in vivo***. In addition to these antitumor effects, THC and CBD also **protect non-transformed central nervous system cells**, particularly astrocytes, from cytotoxic insults.

Here, we analyzed the effects of combined cannabinoid administration (THC:CBD 1:1) and RT on GICs and the glioma microenvironment.

Our results show that **THC:CBD (1:1) enhances the inhibitory effects of RT** on: (i) GICs proliferation *in vitro*; and (ii) the ability of GICs to generate murine orthotopic xenografts. Likewise, ***in vivo* treatment with RT + THC:CBD (1:1) decreased the growth of tumor xenografts generated with GICs and enhanced the survival of the animals** at a higher extent than individual treatments. Cannabinoids also protected primary astrocyte cultures from RT-induced cytotoxicity. Moreover, THC:CBD (1:1) abolished the ability of conditioned medium from irradiated astrocytes to promote GICs self-renewal capacity and radioresistance. Finally, using **CRISPR/Cas9 library screening** together with **RNA-seq datasets from GBM patients**, we identified candidate **genes** associated with **sensitivity or resistance to cannabinoid + RT treatment**.

Altogether, these findings support **cannabinoid administration as a potential strategy to increase GICs sensitivity to radiotherapy while reducing RT-induced damage in non-transformed glial cells**.



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